Topic: Systems of higher order equations

Question: Solve the system of differential equations.

$$x_1' = x_1 + 4x_3$$

$$x_2' = 2x_2$$

$$x_3' = 3x_1 + x_2 - 3x_3$$

Answer choices:

A $\vec{x} = c_1 \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{4t} + c_3 \begin{bmatrix} 4 \\ -1 \\ 6 \end{bmatrix} e^{5t}$

B $\vec{x} = c_1 \begin{bmatrix} 7 \\ -4 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} e^{-5t}$

C $\vec{x} = c_1 \begin{bmatrix} 8 \\ 3 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 9 \\ 8 \\ 7 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix} e^{-5t}$

D $\vec{x} = c_1 \begin{bmatrix} 4 \\ -7 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} e^{3t} + c_3 \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix} e^{-5t}$

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Solution: D

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 3 & 1 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 3 & 1 & -3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 0 & 4 \\ 0 & 2-\lambda & 0 \\ 3 & 1 & -3-\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 4 \\ 0 & 2-\lambda & 0 \\ 3 & 1 & -3-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)\begin{vmatrix} 2-\lambda & 0 \\ 1 & -3-\lambda \end{vmatrix} + 4\begin{vmatrix} 0 & 2-\lambda \\ 3 & 1 \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)(2-\lambda)(-3-\lambda) + 4(2-\lambda)(-3)$$

$$|A - \lambda I| = -6 + 7\lambda - \lambda^3 - 24 + 12\lambda$$

$$|A - \lambda I| = -\lambda^3 + 19\lambda - 30$$

$$|A - \lambda I| = (2-\lambda)(3-\lambda)(-5-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

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$$(2-\lambda)(3-\lambda)(-5-\lambda)=0$$

$$\lambda=2,\ 3,\ -5$$

We'll handle $\lambda_1=2$ first, starting by finding $A-2I$.

$$A-2I=\begin{bmatrix}1-2 & 0 & 4\\ 0 & 2-2 & 0\\ 3 & 1 & -3-2\end{bmatrix}=\begin{bmatrix}-1 & 0 & 4\\ 0 & 0 & 0\\ 3 & 1 & -5\end{bmatrix}$$

We get the system of equations

$$-k_1+4k_3=0$$

$$3k_1+k_2-5k_3=0$$

Solving the first equation gives $k_1=4k_3$. Plugging this into the second equation gives

$$3(4k_3)+k_2-5k_3=0$$

$$k_2+7k_3=0$$

$$k_2=-7k_3$$

If we choose $k_3=1$, then the Eigenvector is

$$\overrightarrow{k_1}=\begin{bmatrix}4\\ -7\\ 1\end{bmatrix}$$

and therefore the solution vector is

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$$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t} = \begin{bmatrix} 4 \\ -7 \\ 1 \end{bmatrix} e^{2t}$$

For $\lambda_2 = 3$, we start by finding $A - 3I$.

$$A - 3I = \begin{bmatrix} 1-3 & 0 & 4 \\ 0 & 2-3 & 0 \\ 3 & 1 & -3-3 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 4 \\ 0 & -1 & 0 \\ 3 & 1 & -6 \end{bmatrix}$$

We get the system of equations

$$-2k_1 + 4k_3 = 0$$

$$-k_2 = 0$$

$$3k_1 + k_2 - 6k_3 = 0$$

We can see from the second equation that $k_2 = 0$, and from the first and third equations that $k_1 = 2k_3$. If we choose $k_3 = 1$, then the Eigenvector is

$$\overrightarrow{k_2} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

and therefore the solution vector is

$$\overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} e^{3t}$$

For $\lambda_3 = -5$, we start by finding $A + 5I$.

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$$A+5I=\begin{bmatrix}1+5 & 0 & 4\\ 0 & 2+5 & 0\\ 3 & 1 & -3+5\end{bmatrix}=\begin{bmatrix}6 & 0 & 4\\ 0 & 7 & 0\\ 3 & 1 & 2\end{bmatrix}$$

We get the system of equations

$$6k_1+4k_3=0$$

$$7k_2=0$$

$$3k_1+k_2+2k_3=0$$

We can see from the second equation that $k_2=0$, and from the first and third equations that $k_1=-(2/3)k_3$. If we choose $k_3=-3$, then the Eigenvector is

$$\overrightarrow{k_3}=\begin{bmatrix}2\\0\\-3\end{bmatrix}$$

and therefore the solution vector is

$$\overrightarrow{x_3}=\overrightarrow{k_3}e^{\lambda_3 t}=\begin{bmatrix}2\\0\\-3\end{bmatrix}e^{-5t}$$

So the general solution to the homogeneous system is

$$\overrightarrow{x}=c_1\overrightarrow{x_1}+c_2\overrightarrow{x_2}+c_3\overrightarrow{x_3}$$

$$\overrightarrow{x}=c_1\begin{bmatrix}4\\-7\\1\end{bmatrix}e^{2t}+c_2\begin{bmatrix}2\\0\\1\end{bmatrix}e^{3t}+c_3\begin{bmatrix}2\\0\\-3\end{bmatrix}e^{-5t}$$

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Topic: Systems of higher order equations

Question: Solve the system of differential equations.

$$x_1' = -x_2$$

$$x_2' = x_1$$

$$x_3' = x_3$$

Answer choices:

A $\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} i \\ 1 \\ -i \end{bmatrix} e^{it} + c_3 \begin{bmatrix} -i \\ 0 \\ 1 \end{bmatrix} e^{-it}$

B $\vec{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} i \\ 1 \\ 0 \end{bmatrix} e^{it} + c_3 \begin{bmatrix} -i \\ 1 \\ 0 \end{bmatrix} e^{-it}$

C $\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 0 \\ 1 \\ i \end{bmatrix} e^{it} + c_3 \begin{bmatrix} 0 \\ -i \\ 1 \end{bmatrix} e^{-it}$

D $\vec{x} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix} e^{it} + c_3 \begin{bmatrix} 1 \\ 1 \\ -i \end{bmatrix} e^{-it}$

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Solution: B

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -\lambda & -1 & 0 \\ 1 & -\lambda & 0 \\ 0 & 0 & 1 - \lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} -\lambda & -1 & 0 \\ 1 & -\lambda & 0 \\ 0 & 0 & 1 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = -\lambda \begin{vmatrix} -\lambda & 0 \\ 0 & 1 - \lambda \end{vmatrix} + 1 \begin{vmatrix} 1 & 0 \\ 0 & 1 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = -\lambda(-\lambda(1 - \lambda)) + 1 - \lambda$$

$$|A - \lambda I| = -\lambda^3 + \lambda^2 - \lambda + 1$$

$$|A - \lambda I| = (\lambda^2 + 1)(1 - \lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(\lambda^2 + 1)(1 - \lambda) = 0$$

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$$\lambda = 1,\ i,\ -i$$

We'll handle $\lambda_1 = 1$ first, starting by finding $A - 1I$.

$$A - 1I = \begin{bmatrix} -1 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1-1 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

We get the system of equations

$$-k_1 - k_2 = 0$$

$$k_1 - k_2 = 0$$

Solving this system gives us $k_1 = k_2 = 0$. If we choose $k_3 = 1$, then the Eigenvector is

$$\overrightarrow{k_1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

and therefore the solution vector is

$$\overrightarrow{x_1} = c_1 \overrightarrow{k_1} e^{\lambda_1 t} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^t$$

For $\lambda_2 = i$, we start by finding $A - iI$.

$$A - iI = \begin{bmatrix} -i & -1 & 0 \\ 1 & -i & 0 \\ 0 & 0 & 1-i \end{bmatrix}$$

We get the system of equations

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$$-ik_1 - k_2 = 0$$

$$k_1 - ik_2 = 0$$

$$(1 - i)k_3 = 0$$

Solving this system gives us $k_1 = ik_2$ and $k_3 = 0$. If we choose $k_2 = 1$, then the Eigenvector is

$$\overrightarrow{k_2} = \begin{bmatrix} i \\ 1 \\ 0 \end{bmatrix}$$

So the solution vector is

$$\overrightarrow{x_2} = c_2\overrightarrow{k_2}e^{\lambda_2 t} = c_2 \begin{bmatrix} i \\ 1 \\ 0 \end{bmatrix} e^{it}$$

For $\lambda_2 = -i$, we start by finding $A + iI$.

$$A - I = \begin{bmatrix} i & -1 & 0 \\ 1 & i & 0 \\ 0 & 0 & 1 + i \end{bmatrix}$$

We get the system of equations

$$ik_1 - k_2 = 0$$

$$k_1 + ik_2 = 0$$

$$(1 + i)k_3 = 0$$

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Solving this system gives us $k_1 = -ik_2$ and $k_3 = 0$. If we choose $k_2 = -1$, then the Eigenvector is

$$\overrightarrow{k_3} = \begin{bmatrix} i \\ -1 \\ 0 \end{bmatrix}$$

So the solution vector is

$$\overrightarrow{x_3} = c_3\overrightarrow{k_3}e^{\lambda_3 t} = c_3 \begin{bmatrix} i \\ -1 \\ 0 \end{bmatrix} e^{-it}$$

So the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t} + c_2 \begin{bmatrix} i \\ 1 \\ 0 \end{bmatrix} e^{it} + c_3 \begin{bmatrix} i \\ -1 \\ 0 \end{bmatrix} e^{-it}$$

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Topic: Systems of higher order equations

Question: Solve the system of differential equations.

$$x_1' = x_1 + 2x_2 + 2x_3$$

$$x_2' = x_2 + x_3$$

$$x_3' = x_3$$

Answer choices:

A $\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} e^t \right) + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{t^2}{2} e^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} e^t \right)$

B $\vec{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-t} + c_2 \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ \frac{2}{3} \\ 0 \end{bmatrix} e^{4t} \right) + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{t^2}{2} e^t + \begin{bmatrix} \frac{2}{3} \\ 0 \\ 1 \end{bmatrix} te^t + \begin{bmatrix} \frac{4}{7} \\ -\frac{3}{8} \\ \frac{1}{2} \end{bmatrix} e^t \right)$

C $\vec{x} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} e^t \right) + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ \frac{3}{4} \end{bmatrix} \frac{t^2}{2} e^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 1 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} e^t \right)$

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D $$\vec{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \end{bmatrix} e^t \right) + c_3 \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \frac{t^2}{2} e^t + \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{bmatrix} e^{3t} \right)$$

Solution: A

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 & 2 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 2 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda) \begin{vmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{vmatrix}$$

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$$|A - \lambda I| = (1 - \lambda)^3$$

Solve the characteristic equation for the Eigenvalues.

$$(1 - \lambda)^3 = 0$$

$$\lambda = 1,\ 1,\ 1$$

We'll start by finding $A - 1I$.

$$A - 1I = \begin{bmatrix} 1-1 & 2 & 2 \\ 0 & 1-1 & 1 \\ 0 & 0 & 1-1 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

We get the system of equations

$$2k_2 + 2k_3 = 0$$

$$k_3 = 0$$

So $k_2 = k_3 = 0$, and if we choose $k_1 = 1$, we find only one Eigenvector.

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

and therefore the solution vector is

$$\overrightarrow{x_1} = c_1 \overrightarrow{k_1} e^{\lambda_1 t} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t$$

Because we only find one linearly independent Eigenvector for the one Eigenvalue $\lambda_1 = \lambda_2 = \lambda_3 = 1$, we have to use $\overrightarrow{k_1} = (1,0,0)$ to find a second solution.

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$$(A - \lambda_2 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 0 & 2 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$2p_2 + 2p_3 = 1$$

$$p_3 = 0$$

Solving this system gives $p_2 = 1/2$ and $p_3 = 0$. If we choose $p_1 = 0$, then the Eigenvector is

$$\overrightarrow{p_1} = \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

and therefore the solution vector is

$$\overrightarrow{x_2} = \overrightarrow{k_1} te^{\lambda_1 t} + \overrightarrow{p_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} e^t$$

But we still need another solution. We'll use $\overrightarrow{p_1} = (0,1/2,0)$ to find a third solution.

$$(A - \lambda_1 I)\overrightarrow{q_1} = \overrightarrow{p_1}$$

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$$\begin{bmatrix} 0 & 2 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

Expanding this matrix equation as a system of equations gives

$$2q_2 + 2q_3 = 0$$

$$q_3 = \frac{1}{2}$$

Solving this system gives $q_2 = -1/2$ and $q_3 = 1/2$. If we choose $q_1 = 0$, then the Eigenvector is

$$\overrightarrow{q_1} = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

and therefore the solution vector is

$$\overrightarrow{x_3} = \overrightarrow{k_1}\frac{t^2}{2}e^{\lambda_1 t} + \overrightarrow{p_1}te^{\lambda_1 t} + \overrightarrow{q_1}e^{\lambda_1 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\frac{t^2}{2}e^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix}te^t + \begin{bmatrix} 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}e^t$$

So the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

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$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t + c_2 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} e^t \right) + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{t^2}{2} e^t + \begin{bmatrix} 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} te^t + \begin{bmatrix} 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} e^t \right)$$

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